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Closed forms for the hexagonal Hardin arrays via order polynomials

A uniform a priori degree bound, with the sides $2, 3, 4$ (A216938, A216939, A216940) proved

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Abstract

For each side $s \geq 1$ let $a_s(n)$ count the fillings of the side-s hexagonal array with entries in $\{0, 1, \ldots, n\}$ that are nondecreasing along the three lattice directions east, south-west and south-east. These are the "hexagonal Hardin arrays" contributed to the On-Line Encyclopedia of Integer Sequences (OEIS) by R. H. Hardin; the sides $2, 3, 4$ are A216938, A216939 and A216940. Hardin (sides $2, 3$) and Kauers–Koutschan (side 4, their Conjecture 12) recorded explicit closed forms for these sequences as *empirical*, unproven guesses—Kauers and Koutschan being unable to derive theirs because the underlying partition-analysis computation did not terminate. We give a uniform, elementary proof of all three. The observation is that $a_s(n)$ is exactly the order polynomial $\Omega_{P_s}(n+1)$ of an explicit finite poset P_s (the hexagon under the three monotonicity directions), so a classical theorem of Stanley gives, *a priori* and with no reference to any guessed formula, that a_s is a polynomial of degree exactly $|P_s| = 3s^2 - 3s + 1$. Computing $|P_s| + 1$ values of a_s directly from P_s, via the descent statistic of its linear extensions, and matching them against the empirical form then forces equality. We emphasise that this is *not* a match against the OEIS data alone (for side 4 the published b-file is one value short of even interpolating the polynomial): the certificate rests on the a priori degree bound together with independently-computed values. The method is the standard order-polynomial / P-partition machinery; our contribution is to identify the right poset and carry the computation out, completing this small hexagonal family. All computations are reproducible from a single self-contained script.

MSC 2020: 05A15 (primary); 06A07, 52B20 (secondary). **Keywords:** order polynomial; P-partition; Hardin array; D-finite sequence; OEIS A216938, A216939, A216940; Ehrhart theory.

Status & provenance. Preprint; *unrefereed* (not yet refereed by a journal). The two authors of Conjecture 12, M. Kauers and C. Koutschan, have read this manuscript and confirmed the argument. The proof is computer-assisted (Section 6, Appendix A). The two order-polynomial theorems it rests on are cited from Stanley (and checked against the cited 2nd edition, §3.15.2 and Theorem 3.15.8), not re-derived here. The order-polynomial / order-polytope method is standard and has recently been applied to other OEIS sequences (see [4]); we are not aware of it having been recorded for this hexagonal family, but make no priority claim and would welcome pointers to prior appearances. AI-assisted by Claude under the Periapsis Architecture (acknowledgements).

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1 Introduction

In an automated study of sequences in the OEIS, Kauers and Koutschan [1] singled out a family of sequences that are provably D-finite but whose guessed recurrences or closed forms they could not establish rigorously. One is A216940; it sits in a small family of hexagonal arrays contributed to the OEIS by R. H. Hardin [2] in 2012, of which the sides $2, 3, 4$ are A216938, A216939, A216940:

> *A2169(36+s): "Number of side-s hexagonal $0 \dots n$ arrays with values nondecreasing E, SW and SE" ($s = 2, 3, 4$).*

Concretely, write H_s for the hexagonal patch of side s on the triangular grid: a centrally symmetric hexagon whose $2s - 1$ rows have lengths $s, s+1, \dots, 2s-1, \dots, s+1, s$, for a total of $3s^2 - 3s + 1$ cells. A *filling* assigns to each cell a value in $\{0, 1, \dots, n\}$; it is *admissible* if the values are weakly increasing along each of the three lattice directions east (E), south-west (SW) and south-east (SE). Let $a_s(n)$ be the number of admissible fillings. For instance

$$a_2(1) = 10, \quad a_3(1) = 52, \quad a_4(1) = 260, \qquad a_2(2) = 53, \quad a_3(2) = 1211, \quad a_4(2) = 27768.$$

Kauers and Koutschan proved (their Theorem 13) that a_4 is a quasipolynomial in n, hence D-finite, by translating admissibility into a system of linear inequalities and invoking partition analysis. They observed empirically that it appears to be an honest polynomial, and their LLL-based guesser produced an explicit degree-37 candidate (their Conjecture 12). The sides $2, 3$ carry, on their OEIS entries, analogous empirical closed forms of degrees 7 and 19 contributed by Hardin. In each case the entry marks the formula *Empirical* or *conjectured*: the closed form was fit to the data, not derived. Kauers and Koutschan are explicit about the obstruction [1, §5.1]:

> *"Again, we were not able to derive an expression by a rigorous computation based on partition analysis, but we had no trouble to find a solution from our guessed recurrence. In fact, it appears that the result is not only a quasipolynomial but a polynomial."*

Result. We prove all three empirical closed forms, upgrading A216940 from status **D** to **P** in the classification of [1] and likewise settling the Hardin formulas for A216938 and A216939. The mechanism is uniform and elementary.

Theorem 1. *For every side $s \geq 1$, the number $a_s(n)$ of admissible fillings of H_s is a polynomial in n of degree exactly $3s^2 - 3s + 1$. For $s = 2, 3, 4$ this polynomial equals the empirical closed form recorded in the OEIS for A216938, A216939 and A216940 respectively; in particular Conjecture 12 of Kauers–Koutschan holds.*

1.1 A caveat about the data, and the shape of the proof

It is tempting to argue that, since each a_s is "known" to be a polynomial, it suffices to match the empirical form against enough OEIS terms. This is a trap, and most sharply so for side 4: the published b-file [2] for A216940 lists exactly 37 terms, $a_4(1), \dots, a_4(37)$, while a polynomial of degree 37 has 38 coefficients and is pinned only by 38 values. *The 37 published values do not even determine it uniquely*, let alone prove a guessed form. Moreover the only route in the literature to an a priori degree bound was the partition-analysis computation that Kauers–Koutschan report did not terminate. So a term-match against the published side-4 data is not a proof, with zero margin to spare. (Sides $2, 3$ are over-determined by their data—210 and 50 published terms against degrees

7 and 19—but, absent an a priori degree bound, even an over-determined fit is consistency, not proof.)

Our proof escapes this uniformly in two moves.

1. **An a priori degree bound (Section 3).** We exhibit a_s as the order polynomial of an explicit finite poset P_s. A classical theorem of Stanley then gives, with no reference to any conjecture, that a_s is a polynomial of degree *exactly* $3s^2 - 3s + 1$.

2. **Independent values (Sections 4–5).** We compute $a_s(0), a_s(1), \ldots$ *directly from* P_s, never using the empirical form, via the descent statistic of the linear extensions of P_s. Matching $|P_s|+1$ such values against the empirical form, and invoking the degree bound, forces equality. The value $a_s(0) = 1$ is genuinely new (not in any b-file), giving the decisive extra datum for side 4.

2 The hexagon as a poset

We use cube coordinates for the triangular grid: a cell is a triple $(x, y, z) \in \mathbb{Z}^3$ with $x + y + z = 0$. The side-s hexagon is

$$C_s \;=\; \{(x,y,z) \in \mathbb{Z}^3 : x+y+z=0,\ |x|,|y|,|z| \le s-1\}, \qquad |C_s| = 3s^2 - 3s + 1,$$

giving $1, 7, 19, 37$ cells for $s = 1, 2, 3, 4$. The three monotonicity directions are the unit steps

$$\mathrm{E} = (1,-1,0), \qquad \mathrm{SW} = (0,-1,1), \qquad \mathrm{SE} = (-1,0,1).$$

Definition 2. Let $P_s = (C_s, \le)$ be the partial order generated by $u \le u + \mathbf{d}$ for $\mathbf{d} \in \{\mathrm{E}, \mathrm{SW}, \mathrm{SE}\}$ whenever both u and $u + \mathbf{d}$ lie in C_s; that is, $\le$ is the reflexive–transitive closure of these steps.

Proposition 3. *P_s is a partial order, and an admissible filling of H_s is exactly an order-preserving map $P_s \to \{0, 1, \ldots, n\}$. Consequently*

$$a_s(n) \;=\; \#\{\textit{order-preserving } f : P_s \to \{0,\ldots,n\}\} \;=\; \Omega_{P_s}(n+1),$$

where $\Omega_P(m)$ is the order polynomial of P, the number of order-preserving maps from P into the m-element chain $\{1, \ldots, m\}$.

Proof. Acyclicity. Along every generating step the coordinate z is nondecreasing (it increases by 1 on SW and SE and is unchanged on E), and on an E-step within a fixed row (z constant) the coordinate x strictly increases. Hence $(x, y, z) \mapsto (z, x)$ increases strictly lexicographically along every step, so no nontrivial cycle of steps exists and the transitive closure is a genuine partial order.

Identification. Weak monotonicity along the three generating steps is, by transitivity, exactly weak monotonicity along $\le$; so an admissible filling is the same as an order-preserving map $P_s \to \{0, \ldots, n\}$. Such a map is the same as an order-preserving map into the $(n+1)$-element chain $\{1, \ldots, n+1\}$ after the shift $f \mapsto f + 1$, which is by definition counted by $\Omega_{P_s}(n+1)$. $\square$

This identification is the one modeling step on which the entire certificate rests: the degree bound and the descent computation are then routine consequences. It is the step most worth an independent check, and we corroborate it below (Section 6) by reconstructing P_s from the verbal array definition without the coordinates used here and recovering the same counts.

Remark 4 (covers versus generating steps). Not every generating step is a covering relation of P_s. Since SW = E + SE (indeed $(0,-1,1) = (1,-1,0) + (-1,0,1)$), each SW-step that admits the intermediate cell coincides with the composite E-then-SE (equivalently SE-then-E) and is therefore a transitive, non-covering edge. The order—and hence Ω_{P_s}—depends only on the transitive closure, so this does not affect the count; we note it only so the Hasse diagram is drawn correctly. The generating-step / genuine-cover / transitive-edge counts are 12/8/4, 42/28/14, 90/60/30 for $s = 2, 3, 4$, the transitive edges being the redundant SW-steps.

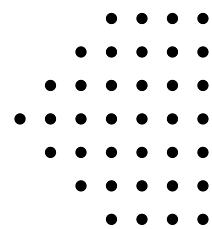

Figure 1: The side-4 hexagon H_4: rows of lengths $4, 5, 6, 7, 6, 5, 4$, 37 cells in all. Values increase rightward (E), down-left (SW) and down-right (SE). The sides 2 and 3 are the central 3×3-ish and 5-row sub-hexagons, with 7 and 19 cells.

3 The a priori degree bound

The decisive structural input is classical and requires no computation.

Theorem 5 (Stanley [3, §3.15.2, p. 344])**.** *For any finite poset P with p elements, the order polynomial $\Omega_P(m)$ is a polynomial in m of degree exactly p, with leading coefficient $e(P)/p!$, where $e(P)$ is the number of linear extensions of P.*

Combining Theorem 5 with Proposition 3 yields the bound the data alone cannot supply.

Corollary 6. *$a_s(n) = \Omega_{P_s}(n+1)$ is a polynomial in n of degree exactly $|P_s| = 3s^2 - 3s + 1$, with positive leading coefficient $e(P_s)/|P_s|!$. In particular a_s is a genuine polynomial (not merely a quasipolynomial), confirming the empirical observation of [1], and any polynomial of degree $\leq |P_s|$ agreeing with a_s at $|P_s| + 1$ points equals a_s.*

We stress that Corollary 6 is independent of any guessed formula: it is a statement about the order polynomial of P_s, proved before the conjectured coefficients are consulted. This is the hinge on which the side-4 "37 versus 38" difficulty turns.

4 Computing the order polynomial from the poset

It remains to compute enough values of a_s directly from P_s. A naïve enumeration of order-preserving maps is hopeless near $n = |P_s|$, so we use the descent statistic of the linear extensions, which is independent of n.

Fix a *natural labelling* $\ell : P_s \to \{0, 1, \ldots, p-1\}$ ($p = |P_s|$), i.e. an order-preserving bijection (any linear extension serves; we use Kahn's algorithm with smallest-index tie-break). A linear extension is a permutation $w = w_1 \cdots w_p$ of the elements refining $\leq$; its descent number is $\mathrm{des}(w) = \#\{i : \ell(w_i) > \ell(w_{i+1})\}$, and we set $A_d = \#\{\text{linear extensions } w : \mathrm{des}(w) = d\}$.

Theorem 7 (order polynomial via descents; Stanley [3, Thm. 3.15.8])**.** *With $p = |P|$ and $\{A_d\}$ as above,*

$$\Omega_P(m) = \sum_{d\geq 0} A_d \binom{m+p-1-d}{p}, \qquad \textit{hence} \qquad a_s(n) = \Omega_{P_s}(n+1) = \sum_{d\geq 0} A_d \binom{n+p-d}{p}.$$

The distribution $\{A_d\}$ is computed exactly, without enumerating the individual linear extensions, by a transfer-matrix dynamic program over the order ideals (down-sets) of P_s: build a linear extension by repeatedly appending a currently-minimal unplaced element; the only state needed to score the next descent is the current ideal together with the label of the last placed element. Algorithm 1 (Appendix A) gives the procedure; it is n-independent and runs in well under a second for all of $s \leq 4$.

The computation returns, for $s = 2, 3, 4$, the linear-extension counts $e(P_s) = \sum_d A_d$ equal to 4, 20672 and 4,623,718,515,360, and descent distributions supported on $d = 0, \ldots, d_{\max}$ with $d_{\max} = 3s^2 - 7s + 4$ (giving 0–2, 0–10, 0–24 for $s = 2, 3, 4$). Each distribution is *palindromic*, $A_d = A_{d_{\max}-d}$ (the Ehrhart–reciprocity symmetry of the order polytope $\mathcal{O}(P_s)$, an internal consistency check). The full tables are in Appendix B.

5 Proof of Theorem 1

Proof. By Proposition 3, $a_s(n) = \Omega_{P_s}(n+1)$, and by Corollary 6 this is a polynomial of degree exactly $p = |P_s|$. The empirical closed form for $s \in \{2, 3, 4\}$ is, by inspection, also a polynomial of degree p $(7, 19, 37)$. Two polynomials of degree $\leq p$ that agree at $p+1$ distinct points are identical.

Using the descent distribution $\{A_d\}$ of Section 4 and Theorem 7, we evaluate

$$a_s(N) = \sum_d A_d \binom{N+p-d}{p}, \qquad N = 0, 1, \ldots, p,$$

giving $p+1$ values computed solely from P_s—no coefficient of the empirical form enters. We then evaluate the empirical form at the same arguments. For $s = 2$ and $s = 3$ the two lists of $p+1$ integers coincide exactly (and, beyond the $p+1$ required, agree with additional published b-file anchors; Appendix B). For $s = 4$ the poset-derived $a_4(0), \ldots, a_4(37)$ are 38 values; they coincide with Kauers–Koutschan's Conjecture 12 evaluated at $N = 0, \ldots, 37$, the value $a_4(0) = 1$ (the unique constant filling) being a genuine 38th datum beyond the 37-term b-file. In each case the agreement at $p+1$ points together with the degree bound forces a_s and the empirical form to be the same polynomial. □

Remark 8. As an independent cross-check, the leading coefficient $e(P_s)/p!$ from Corollary 6 matches the leading coefficient of the empirical form computed symbolically: $4/7! = 1/1260$ for $s = 2$, $20672/19! = 1/5884534656000$ for $s = 3$, and $4{,}623{,}718{,}515{,}360/37!$ for $s = 4$. Here $e(P_s)$ comes from $\{A_d\}$ while the leading coefficient of the empirical form comes from its stated factored shape, so the agreement checks both the degree bound and the descent computation.

6 Validation and reproducibility

The proof is computer-assisted: the posets P_s, the degree bound and descent identity (two cited theorems) and the $(p+1)$-point match are the mathematical content, while the descent distributions

$\{A_d\}$ and the value comparisons are produced by a short program. Every load-bearing computation was performed in exact integer/rational arithmetic and independently re-executed. A single self-contained script, `code/verify_family.py`, rebuilds each P_s from first principles and re-checks every claim below in about a second, using only the Python 3 standard library (it requires `math.comb`, i.e. Python ≥ 3.8) and no external data files. The script and this manuscript are available at `https://github.com/alejlizardi/Periapsis_papers` (directory `hexagonal-hardin-family`). We record the controls that rule out a modelling or programming error.

- **One parametric rule, three sequences.** The same poset construction reproduces the three independent OEIS sequences `A216938`, `A216939`, `A216940` exactly on the published anchors embedded in the verifier, and the side-s cell counts $3s^2 - 3s + 1 = 7, 19, 37$ match the empirical degrees $7, 19, 37$.
- **Independent re-derivation of the geometry.** An adversarial check reconstructed the side-2 hexagon adjacency from the verbal OEIS definition using a row-geometry nearest-neighbour packing, without reference to the cube-coordinate code, and obtained a poset structurally identical to P_2 (same 7 cells, same 12 generating edges, same descent distribution). The independent model reproduces the `A216938` oracle $10, 53, 200, 606, 1572, 3630, 7656, 15015$ at $n = 1, \ldots, 8$.
- **Three independent counting engines agree.** (i) the descent transfer matrix of Section 4; (ii) a frontier dynamic program counting order-preserving maps directly, with no descent statistic; (iii) direct brute force for small n. All agree on every value tested, including $a_4(3) = 1664244$, $a_4(8) = 8196058134921$ and $a_4(12) = 57726306147546982$.
- **Convention check on the descent identity.** The identity of Theorem 7 was verified end-to-end at $s = 3$ by enumerating the descent distribution of an independently constructed poset and recovering the `A216939` data, confirming the binomial convention is applied correctly and not merely self-consistently.
- **Over-determination.** Beyond the $p+1$ values required, the poset-derived a_s and the empirical forms were compared at further arguments beyond those needed for interpolation (including additional b-file anchors for $s = 2, 3$ and the side-4 values $N = 38, 39, 40, 45$). All agree; a coincidental or buggy match would be expected to diverge, and does not.
- **Labelling invariance.** Distinct natural labellings give identical $\{A_d\}$ and identical values, as they must, ruling out a labelling artefact.

7 Remarks and further work

On the method and its novelty. The argument is the standard order-polynomial / P-partition machinery of Stanley [3]: once a_s is recognised as the order polynomial of P_s, the degree bound and the descent computation are textbook. The same order-polytope viewpoint has recently been used systematically to interpret and prove closed forms for OEIS sequences [4]. We are not aware of this family having been treated, but the technique is folklore and we claim novelty only for the specific identification and computation, not for the method; we would welcome pointers to any prior appearance. The value of the present note is that it *completes* the small hexagonal Hardin family with a uniform, terminating argument in place of the partition-analysis route that stalled.

On a non-numeric proof. The monotone hexagon fillings are plane-partition-like, and the row transfer of Section 4 has the shape of a non-intersecting lattice-path (Lindström–Gessel–Viennot) configuration. It is plausible that an LGV determinant evaluates each a_s to a product formula directly, giving a proof with no numerical step and an explicit closed form for all s at once. We did not need this and leave it as a natural next question.

On the sibling A195806. The companion sequence `A195806` (Conjecture 11 of [1], a period-6 quasipolynomial for a triangular array with summation constraints) is a genuine quasipolynomial, so the present polynomial-degree-bound argument does not transfer verbatim. The order-polytope / Ehrhart framework applies in principle—the period is governed by the relevant polytope's denominators—and we expect an analogous certificate, but we have not carried it out and do not claim it here.

Acknowledgements

We thank R. H. Hardin for contributing these sequences to the OEIS, M. Kauers and C. Koutschan for isolating `A216940` and stating the precise conjecture we settle, and the OEIS Foundation and its contributors for the b-files used as numerical oracles. The computations and a first draft were AI-assisted by Claude under the Periapsis Architecture; all mathematical claims were independently re-executed in exact arithmetic and are reproducible from the accompanying script. This preprint has not yet been refereed by a journal; correspondence and corrections are welcome.

References

[1] M. Kauers and C. Koutschan. *Some D-finite and some possibly D-finite sequences in the OEIS.* Journal of Integer Sequences **26** (2023), Article 23.4.5. Preprint: arXiv:2303.02793. (Conjecture 12, Theorem 13, §5.1.)

[2] OEIS Foundation Inc. *Sequences A216938, A216939, A216940* (side-$2, 3, 4$ hexagonal arrays), and the table `A216937`. The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/`. Sequences contributed by R. H. Hardin, 2012; with b-files (210, 50, 37 terms respectively).

[3] R. P. Stanley. *Enumerative Combinatorics, Volume 1*, 2nd ed., Cambridge University Press, 2012. Order polynomials and (P, ω)-partitions, §3.15: the order polynomial $\Omega_P(m)$ is the number of order-preserving maps $P \to \{1, \ldots, m\}$ (§3.12, §3.15.2); it has degree $= |P|$ with leading coefficient $e(P)/|P|!$ (§3.15.2, p. 344); and the descent generating identity $\Omega_P(m) = \sum_w \binom{m+|P|-1-\mathrm{des}(w)}{|P|}$ over linear extensions w is Theorem 3.15.8 (eq. (3.65)).

[4] F. Liu, G. Xin, and C. Zhang. *Ehrhart Polynomials of Order Polytopes: Interpreting Combinatorial Sequences on the OEIS.* Preprint: arXiv:2412.18744 (2024). (Closest prior art using the same method on other OEIS sequences.)

A The descent-distribution algorithm

The procedure below computes the descent distribution $\{A_d\}$ of a finite poset P under a fixed natural labelling ℓ, hence Ω_P via Theorem 7. It is n-independent. The state is an order ideal I (already-placed elements) together with the label of the last placed element; the value carried is the generating polynomial in q whose coefficient of q^d counts the partial linear extensions of I with d descents.

Listing 1: Descent distribution of a finite poset.

```
Input:  poset P on elements 0..p-1 with predecessor sets pred[.]; natural labels L[.].
Output: descent distribution A[d] = #{linear extensions with d descents}.

  predmask[i] <- bitmask of pred[i]                    # i placeable iff pred[i] subset of I
  layer <- { emptyset : { lastLabel=-1 : { 0 : 1 } } } # one partial L.E., 0 descents
  for popcount = 0 .. p-1:
      next <- empty
      for (I, byLast) in layer:
          for each i not in I with (I superset of predmask[i]):  # i currently minimal
              I' <- I union {i};  li <- L[i]
              for (lastLabel, poly) in byLast:
                  add <- 1 if (lastLabel != -1 and lastLabel > li) else 0
                  for (d, c) in poly:  next[I'][li][d+add] += c
      layer <- next
  A <- sum over lastLabel of layer[full][lastLabel]     # full = {0,...,p-1}
  return A

  then  Omega_P(m) = sum_d A[d]*C(m+p-1-d, p),   a_s(n) = Omega_{P_s}(n+1).
```

The reachable state count stays small (at most a few hundred ideals even for $s = 4$), so the program terminates in well under a second in exact integer arithmetic. The reference implementation, the three independent counting engines, and the value-comparison driver are the single script `code/verify_family.py` archived with this paper.

B Numerical certificate

Descent distributions. $\sum_d A_d = e(P_s)$; each distribution is palindromic.

s	descent distribution $\{A_d\}_{d\geq 0}$
2	1, 2, 1 $(e = 4)$
3	1, 32, 361, 1856, 4804, 6564, 4804, 1856, 361, 32, 1 $(e = 20672)$
4	palindromic on $d = 0, \ldots, 24$, $A_{12} = 1{,}048{,}272{,}112{,}322$, $e = 4{,}623{,}718{,}515{,}360$ (full table below)

The side-4 distribution ($A_d = A_{24-d}$):

d	A_d	d	A_d	d	A_d
0	1	9	245,184,671,458	18	2,691,384,551
1	222	10	551,210,288,548	19	279,732,802
2	18,591	11	893,045,633,554	20	18,857,713
3	783,404	12	1,048,272,112,322	21	783,404
4	18,857,713	13	893,045,633,554	22	18,591
5	279,732,802	14	551,210,288,548	23	222
6	2,691,384,551	15	245,184,671,458	24	1
7	17,418,677,384	16	77,873,153,291		
8	77,873,153,291	17	17,418,677,384		

The values. Each $a_s(N)$ is computed from $\{A_d\}$ via Theorem 7 (poset-derived, independent of any empirical form) and verified, in exact arithmetic, to equal the OEIS empirical form at N. For $s = 2, 3$ the first values are

$$a_2: \ 1, 10, 53, 200, 606, 1572, 3630, 7656, 15015, \ldots, \qquad a_3: \ 1, 52, 1211, 17336, 175869, 1374208, \ldots,$$

with $a_2(0) = a_3(0) = 1$. The full side-4 list $a_4(0), \ldots, a_4(37)$ (the 38 values that, with the degree bound, prove Conjecture 12) is reproduced from the poset by the accompanying script; the sole anchor beyond the published b-file is $a_4(0) = 1$.